DTN over KISS: Regulatory Compliance

DTN over KISS: Regulatory Compliance with OFCOM Amateur Radio Licence Conditions

Executive Summary

AMSAT-UK proposes to conduct experimental transmissions using Delay-Tolerant Networking (DTN) protocols over standard 1200 baud VHF amateur packet radio links. DTN is a suite of networking protocols developed by NASA and the IETF for reliable communications over links with long delays and intermittent connectivity, such as deep-space and satellite links. AMSAT-UK seeks to evaluate these protocols on terrestrial amateur radio as a stepping stone toward their use on future amateur satellite missions.

The system uses standard amateur radio equipment (Yaesu FT-817 transceivers and Mobilinkd TNC3 terminal node controllers) operating within OFCOM-permitted frequency bands and power levels. The software is open-source and implements two protocol modes: a conventional AX.25 packet mode and a Licklider Transmission Protocol (LTP) mode that provides reliable data transfer with acknowledgment and retransmission.

Station identification is maintained at all times in compliance with the OFCOM Amateur Radio Licence (2024 edition). In AX.25 mode, the operator's callsign is embedded in every transmitted frame. In LTP mode, DTN endpoint identifiers containing the operator's callsign (e.g. dtn://g4dpz-1) are used for addressing, supplemented by periodic AX.25 identification beacons to ensure the station remains identifiable to conventional monitoring equipment.

All transmissions are conducted for the purposes of self-training and technical investigation, in accordance with the terms of the amateur radio licence and ITU Radio Regulations Article 25.

Document Purpose

This document is a proposal from AMSAT-UK describing an experimental amateur radio data communications system that implements Delay-Tolerant Networking (DTN) protocols over 1200 baud VHF packet radio. It explains how the system conforms to OFCOM amateur radio licence conditions regarding station identification and is intended for presentation to OFCOM to demonstrate regulatory awareness and compliance.

Proposing Organisation

AMSAT-UK is the Radio Amateur Satellite Corporation of the United Kingdom, a registered charity and membership organisation dedicated to the advancement of amateur radio satellite communications, education, and technical research. AMSAT-UK members have a long history of designing, building, and operating amateur radio satellites and ground station systems, and of working constructively with OFCOM on spectrum matters relating to the amateur satellite service.

This proposal arises from AMSAT-UK's interest in applying Delay-Tolerant Networking techniques — originally developed for space communications — to terrestrial amateur radio links, with the goal of building practical experience and tools that can be applied to future amateur satellite missions.

System Overview

The system implements a layered protocol stack for reliable data transfer between amateur radio stations over RF links characterised by long propagation delays and intermittent connectivity. The hardware consists of standard amateur radio equipment (Yaesu FT-817 transceivers) connected to Mobilinkd TNC3 terminal node controllers via USB, driven by custom open-source software running on Linux computers (Ubuntu laptops and Raspberry Pi).

The protocol stack, from lowest to highest layer:

1. RF physical layer: 1200 baud AFSK on VHF amateur bands
2. KISS framing: host-to-TNC serial protocol for packet delineation
3. AX.25 UI frames (current) / LTP segments (planned): data link / transport
4. Application data: user messages, ping payloads, or DTN bundles

Applicable Regulations and Standards

OFCOM Regulations

- OFCOM Amateur Radio Licence Conditions Booklet (OFW611), February 2024 edition. This is the primary regulatory document governing amateur radio operation in the United Kingdom. (Source: OFCOM)

- OFCOM Interface Requirement IR 2028: "UK Interface Requirement for Radio Equipment in the Amateur and Amateur Satellite Services." This defines the technical parameters for amateur radio equipment. (Source: OFCOM)

- Wireless Telegraphy Act 2006 (WTA 2006): The primary UK legislation governing the use of radio spectrum. (Source: UK Government)

International Regulations

- ITU Radio Regulations, Article 25: "Amateur Service and Amateur-Satellite Service." Provision 25.1 governs radiocommunications between amateur stations of different countries. Provision 25.6 requires that amateur operators demonstrate operational and technical qualifications. (Source: ITU)

Protocol Standards

- AX.25 Link Access Protocol for Amateur Packet Radio, Version 2.2 (July 1998). Published by the Tucson Amateur Packet Radio Corporation (TAPR). This is the standard amateur radio data link protocol used in the current implementation. (Source: TAPR)

- KISS Protocol: "A Simple Host-to-TNC Communications Protocol," by Mike Chepponis, K3MC, and Phil Karn, KA9Q (August 1986). Published by TAPR. Defines the serial framing protocol between host computer and TNC hardware. (Source: TAPR)

- RFC 5326: "Licklider Transmission Protocol — Specification," by M. Ramadas, S. Burleigh, and S. Farrell (September 2008). Defines the LTP reliable transport protocol for delay-tolerant links. (Source: IETF)

- RFC 5325: "Licklider Transmission Protocol — Motivation," by S. Burleigh, M. Ramadas, and S. Farrell (September 2008). Describes the rationale for LTP over high-delay links. (Source: IETF)

- RFC 9171: "Bundle Protocol Version 7," by S. Burleigh, K. Fall, and E. Birrane (January 2022). Defines the DTN Bundle Protocol for store-and-forward networking. (Source: IETF)

- RFC 4838: "Delay-Tolerant Networking Architecture," by V. Cerf et al. (April 2007). Defines the overall DTN architecture. (Source: IETF)

- CCSDS 734.1-B-1: "Licklider Transmission Protocol (LTP) for CCSDS." The Consultative Committee for Space Data Systems standard for LTP, used by NASA and ESA for deep-space communications. (Source: CCSDS)

Station Identification Compliance

OFCOM Licence Requirement

The OFCOM Amateur Radio Licence (2024 edition) requires that a station must be clearly identifiable at all times during transmission, and that the licensee must identify the station as frequently as practicable using the assigned callsign. Content rephrased for compliance with licensing restrictions.

The 2024 licence revision replaced the previous prescriptive "every 15 minutes" rule with a principles-based approach requiring stations to be clearly identifiable at all times. (Source: Essex Ham)

How This System Complies

Current Implementation (AX.25 Mode)

In the current AX.25-based implementation, every transmitted packet contains the operator's callsign embedded in the AX.25 frame header as both source and destination address fields. This is inherent to the AX.25 protocol design.

Each AX.25 UI frame contains: - Destination address field (7 bytes): the remote station's callsign and SSID - Source address field (7 bytes): the transmitting station's callsign and SSID

For example, a transmission from station G4DPZ-1 to G4DPZ-2 carries both callsigns in every single packet transmitted over the air. The callsigns are encoded in the standard AX.25 address format (characters left-shifted by one bit, per AX.25 v2.2 Section 2.2.13), and are decodable by any standard packet radio monitoring equipment.

This means the station is identified in every transmission, exceeding the OFCOM requirement for identification "as frequently as practicable."

The system's CLI requires the operator to specify their callsign explicitly:

```
./kiss_interface send --device /dev/ttyACM0 --src G4DPZ-1 --dst G4DPZ-2 "message"
```

Planned Implementation (LTP over KISS Mode)

In the planned LTP-over-KISS implementation, the AX.25 framing layer is replaced by LTP segments encapsulated directly in KISS frames. LTP uses numeric Engine IDs rather than callsign strings in its segment headers.

To maintain compliance with OFCOM identification requirements, the system uses DTN endpoint identifiers that embed the operator's callsign:

```
dtn://g4dpz-1
```

The callsign is mapped to a numeric Engine ID using a deterministic hash function. The mapping between DTN endpoint identifier and Engine ID is maintained by both communicating stations, ensuring that any received LTP segment can be traced back to the originating callsign.

Additionally, the system will implement periodic AX.25 identification beacons during LTP sessions. At configurable intervals (default: every 10 minutes), the system will transmit a standard AX.25 UI frame containing the operator's callsign, ensuring that the station remains identifiable to any monitoring station using conventional packet radio equipment. This hybrid approach satisfies the OFCOM requirement that the station be "clearly identifiable at all times" while allowing the use of the more efficient LTP protocol for data transfer.

The CLI requires the operator to specify their DTN endpoint (which contains their callsign):

```
./kiss_interface ltp-send --device /dev/ttyACM0 --local dtn://g4dpz-1 --remote dtn://g4dpz-2 "message"
```

Summary of Identification Mechanisms

Mode	Identification Method	Frequency
AX.25	Callsign in every AX.25 frame header	Every packet

send/receive/echo		
AX.25 ping	Callsign in every AX.25 frame header	Every ping/reply
LTP send/receive	DTN endpoint ID containing callsign + periodic AX.25 beacon	Every session + periodic beacon

Purpose of the Experiment

This system is being developed for the purpose of self-training and technical investigation, which are recognised purposes under the OFCOM amateur radio licence and ITU Radio Regulations Article 25. Specifically:

1. Investigating the application of Delay-Tolerant Networking protocols (originally developed by NASA/JPL for deep-space communications) to terrestrial amateur radio links
2. Evaluating the performance of the Licklider Transmission Protocol over 1200 baud VHF packet radio
3. Measuring round-trip times and link reliability characteristics for constrained RF links
4. Developing open-source tools for the amateur radio community

All transmissions are non-commercial, between licensed amateur radio stations, using standard amateur radio equipment operating within the permitted frequency bands and power levels specified in the OFCOM licence.

Equipment

Component	Description
Transceiver	Yaesu FT-817 (5W VHF/UHF, operating within OFCOM-permitted bands and power levels)
TNC	Mobilinkd TNC3 (USB KISS TNC, 1200 baud AFSK)
Computer	Ubuntu Linux laptop / Raspberry Pi
Software	Custom open-source C tool (`kiss_interface`), source available at https://github.com/g4dpz/ion-dtn-terrestrial

Contact

This proposal is submitted on behalf of AMSAT-UK.

Licensee callsign: G4DPZ Organisation: AMSAT-UK (Radio Amateur Satellite Corporation of the United Kingdom) Website: https://amsat-uk.org

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